

Project: Build a Data Mart in SQL

Finalization Phase – Abstract

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Abstract

This project presents the design and implementation of a comprehensive relational database management system (DBMS) to support core Airbnb-like operations, including user management, property listings, bookings, payments, reviews, and support features. The system is structured around 22 normalized tables, ensuring robust data management, referential integrity, and extensibility for real-world deployment.

Database Design and Implementation: The schema employs the InnoDB storage engine with UTF-8 (utf8mb4) encoding to guarantee internationalization and data integrity. All tables are normalized and interlinked to support complex business logic and data associations.

System Metadata:

- **Total Tables:** 22
- **Total Records:** 440+ (minimum 20 entries per table)
- **Database Size:** Approximately 0.94 MB

Core Components:

- **User Management:** (user_profiles, hosts, guests, admin_users)
- **Property Management:** (property_listings, property_images, amenities, listing_amenities, property_rules, calendar_availability, pricing_history)
- **Transaction Management:** (reservations, transactions, cancellation_policies)
- **Support and Communication:** (property_reviews, host_reviews, messages, support_tickets, user_activity_logs)
- **Referral and Dispute Handling:** (referral_programs, referral_rewards, dispute_resolution)

Key Functionalities:

- Multi-role user management for hosts, guests, and administrators.
- Sophisticated reservation and payment processes, including platform-based payment retention and commission logic.
- Flexible property amenities and rules managed via many-to-many junction tables.
- Integrated review and messaging systems for user feedback and communication.
- Recursive and ternary relationships for referral tracking, administrative hierarchies, and dispute resolution.
- Advanced support for status tracking, cancellation policies, and user activity logging.

System Integration Highlights:

- Robust handling of complex data associations using triple and recursive relationships (`referral_rewards`, `dispute_resolution`, `admin.users`).
- Secure platform payment system holding guest payments for 24 hours post check-in to enhance transaction integrity and satisfaction.
- Comprehensive property management leveraging normalized junction tables for rules and amenities.

Outcome: The resulting system demonstrates technical sophistication and practical utility, providing a solid foundation for scalable, real-world deployment with minimal adaptation required.